

# ALVEOLAR BONE

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# Alveolar bone

## Salient Features

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- **Mineralized connective tissue**
  - To supports and protects the teeth
  - component of periodontium.
- **The significant mechanical properties**
  - Strength
  - Plasticity to permit remodeling as per functional demands.
- **Short and intermittent mechanical stimuli required**
  - to maintain structure and mass and to perform functions.

# **1.What is the chemical composition of alveolar bone?**

**Matrix of bone is a composite material of inorganic mineral and an organic matrix.**

- 1. Inorganic component (60%) is hydroxyapatite –  $\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$ .**
- 2. Organic component (25%) is 90% type I collagen. Remaining 10% is complex mixture of proteins, including growth factors, osteocalcin, osteonectin, osteopontin, glycoproteins produced by osteoblasts, along with type I collagen.**
- 3. About 15% water**

## **2.Differentiate between modeling and remodeling of bone?**

### **Modeling**

- **Bone deposition at site other than bone resorption.**
- **Associated with change in size and shape of bone.**

### **Remodeling**

- **Also known as internal turnover.**
- **Bone deposition at the site of bone resorption.**
- **No change in overall shape.**
- **as bone remodeling occur throughout life, there is a continuous need for stem/osteoprogenitor cells.**

### **3. Define Osteoid?**

- A layer of newly deposited, unmineralized, bone matrix is termed as 'osteoid' of approximately 5–10  $\mu\text{m}$  thickness and secreted by osteoblast.
- Mineralization of osteoid is dependent on the hormonally active form of vitamin D (1,25-dihydroxyvitamin D).

## 4. Classify Bone?

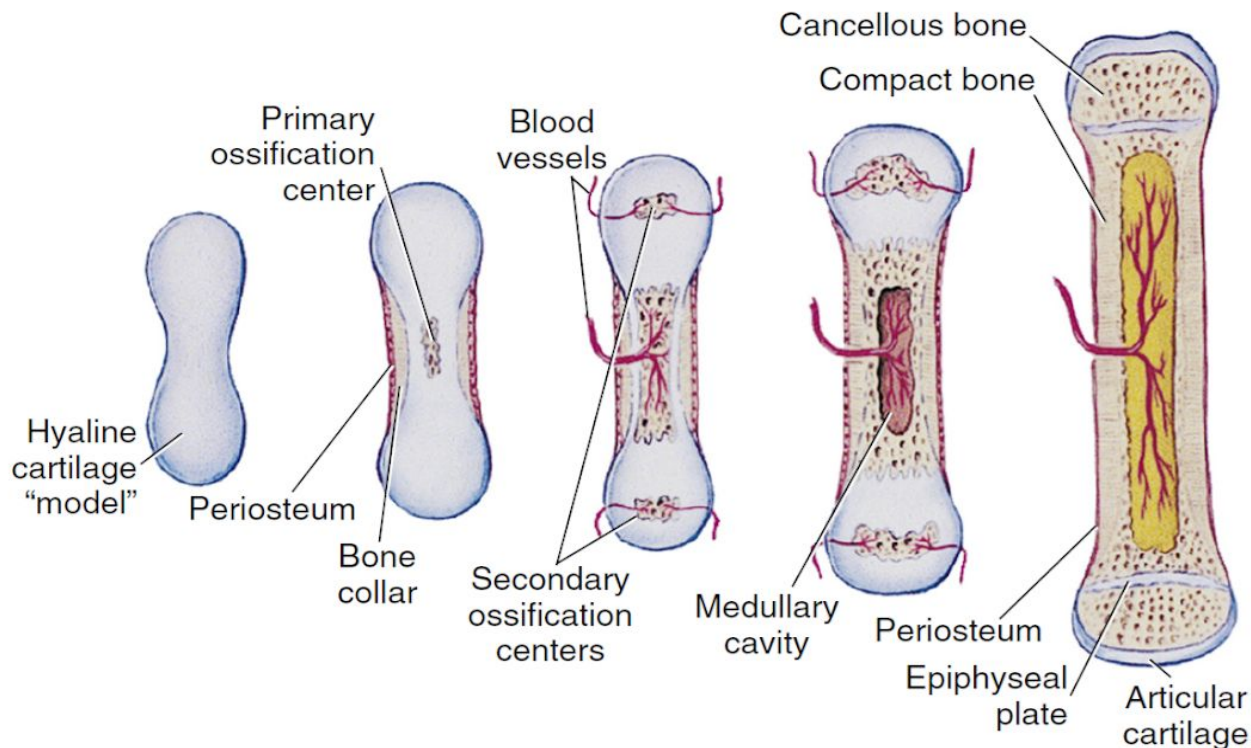
- **Developmentally**-----endochondral and intramembranous bone
- **Histologically**-----compact(cortical) and cancellous(spongy) bone
- **Biologically** -----newly formed, immature(woven) and adult(lamellar) bone

### 3a. Define endochondral ossification

Osteoid forms within a cartilage model followed by osteoblasts penetration in the disintegrating cartilage.

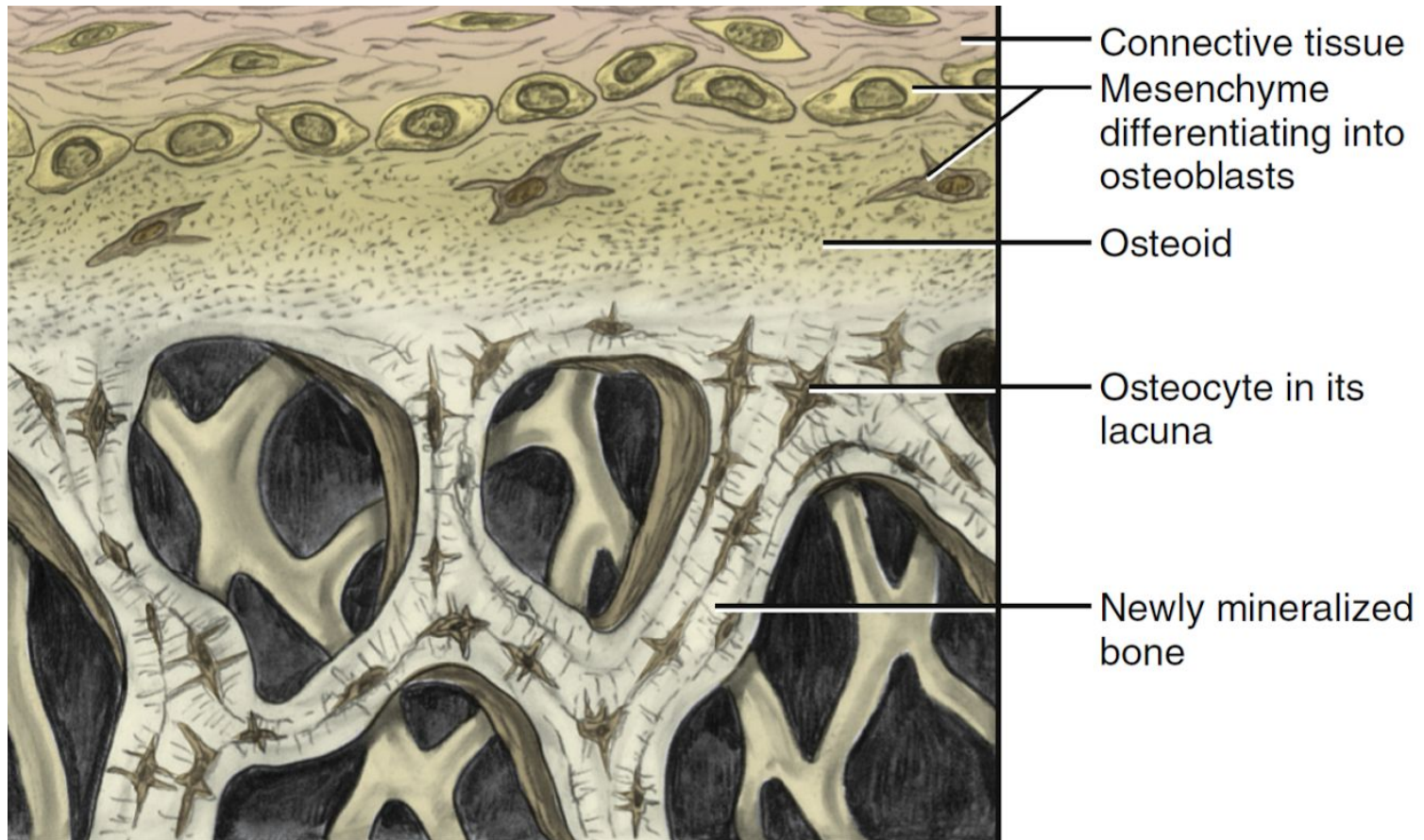
Primary ossification center are formed and osteoid grows toward the ends of the bone and becomes mineralized.

Later, after birth, secondary ossification centers are formed for further growth of bones.



### 3b. Define Intramembranous Bone

**mesenchyme differentiates into osteoblasts to form osteoid, which later matures into bone.**





**Woven bone  
(immature fracture repair)**

**Lamellar bone  
(mature adult)**

**Osteocytes larger and  
irregularly spaced**

**Osteocytes smaller, flattened  
and regularly spaced**

**Randomly oriented collagen  
fibres**

**Collagen fibres arranged in  
regular plywood orientation**

**Enriched in acid  
phosphoproteins and BSP**

**Enriched in osteocalcin**

**Rapid matrix mineralisation**

**Delayed matrix mineralisation**

**Forms rapidly**

**Forms slowly**

**Rapid turnover**

**Slow turnover**

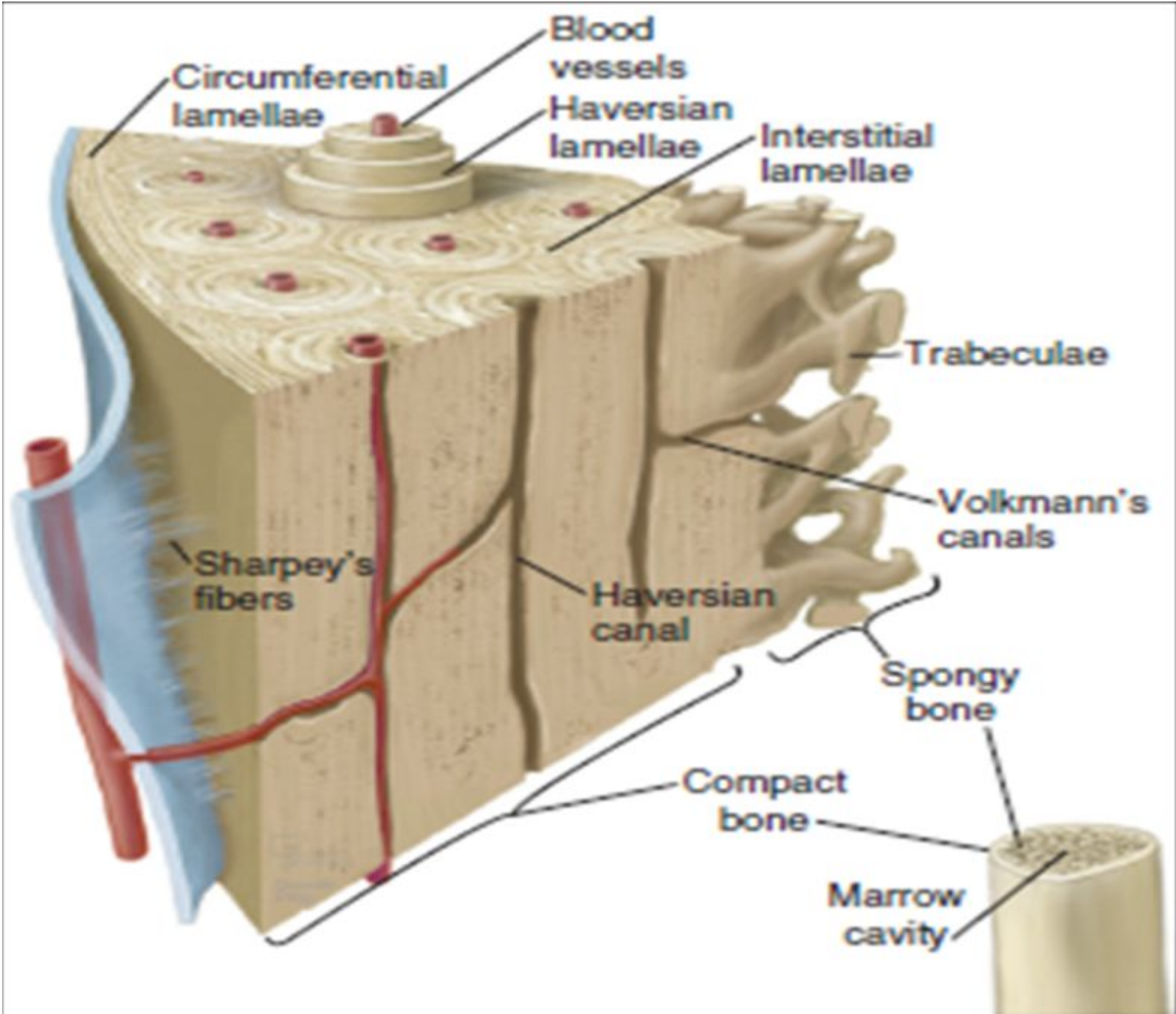
# What are the components of Compact Bone?

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The basic structural unit of bone is the osteon (Haversian system).

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Its components are the central canal (haversian canal) of about  $50\mu\text{m}$  in diameter, together with the about 20 concentric lamellae, with each lamellae of  $3\text{-}5\mu\text{m}$  thickness.



# **What is the structural organization of Compact Bone? Or Differentiate between circumferential lamellae, concentric lamellae and interstitial lamellae?**

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Lamellae arranged in two major patterns in form of external periosteal surface and internal endosteal surface lie in parallel layers completely surrounding the bony surfaces and are known as circumferential lamellae.

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Lamellae arranged as small, concentric layers around a central neurovascular canal are the concentric lamellae.

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After remodeling, lamellae present as fragments of previous osteon between newly deposited osteon are interstitial lamellae.

# **What is the structural organization of Cancellous Bone?**

- lamellae are apposed to each other to form trabeculae up to about 50  $\mu\text{m}$  thick aligned along lines of stress.
- The trabeculae surround the marrow spaces of alveolar bone are thicker and derive their nutrition by diffusion.

| <b>COMPACT BONE</b>                                                              | <b>CANCELLOUS BONE</b>                                                                                                     |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>Dense cortical</b>                                                            | <b>spongy</b>                                                                                                              |
| <b>Arranged in form of circumferential, concentric and interstitial lamellae</b> | <b>Marrow space is red and hematopoietic in young bone and pale and non-hematopoietic with more fat cells in old bone.</b> |
| <b>80% bone of the body</b>                                                      | <b>20% bone of the body</b>                                                                                                |
| <b>Bone remodel 3% of its mass</b>                                               | <b>Bone remodel 25% of its mass</b>                                                                                        |
| <b>Mechanical and protective role</b>                                            | <b>Role in mineral metabolism and structure formation</b>                                                                  |

## Developmental features

There are four main types of bone cells:

- ✓ three (osteoblast, osteocyte and bone-lining cell) mesenchymal in origin
- ✓ while the fourth (osteoclast) is haemopoietic in origin.

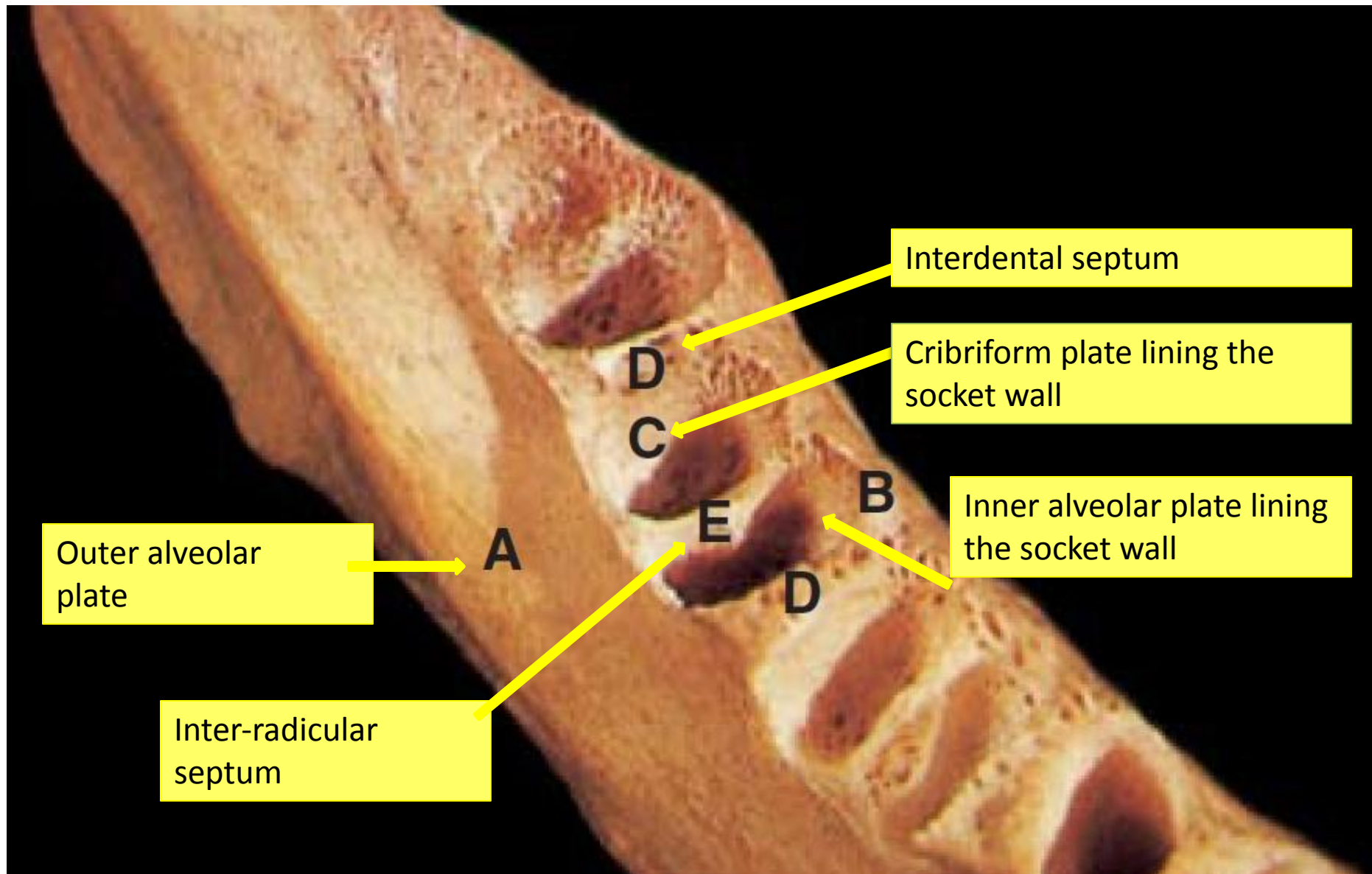


Fig. 13.3 Mandible with teeth removed to demonstrate the components of alveolar bone



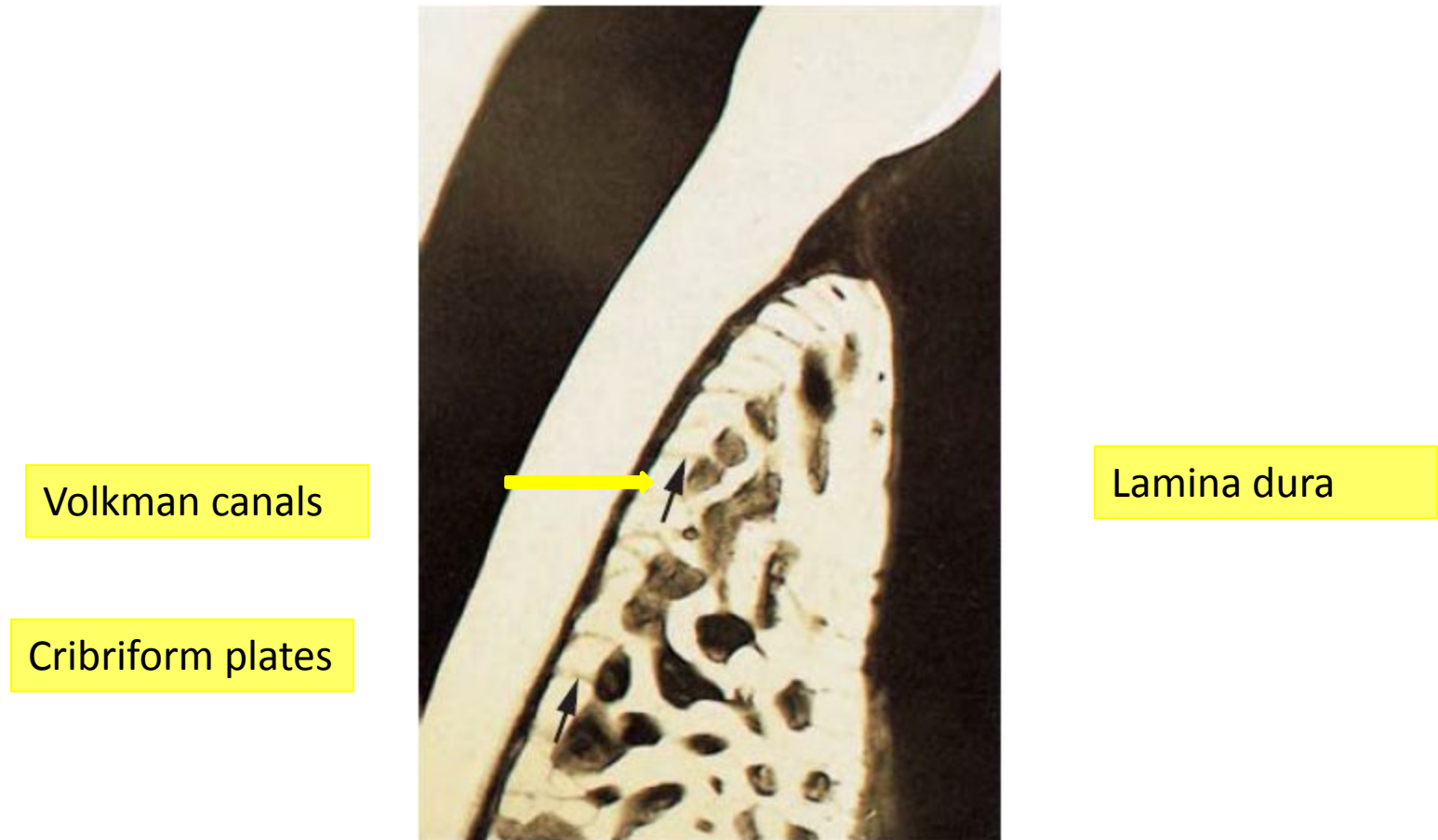
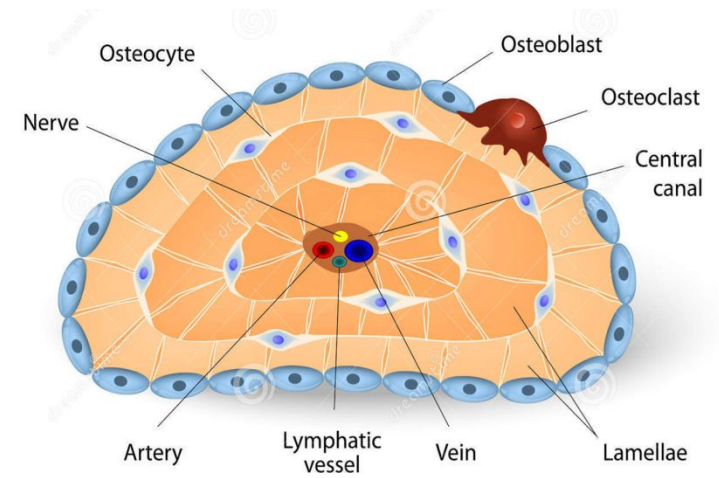
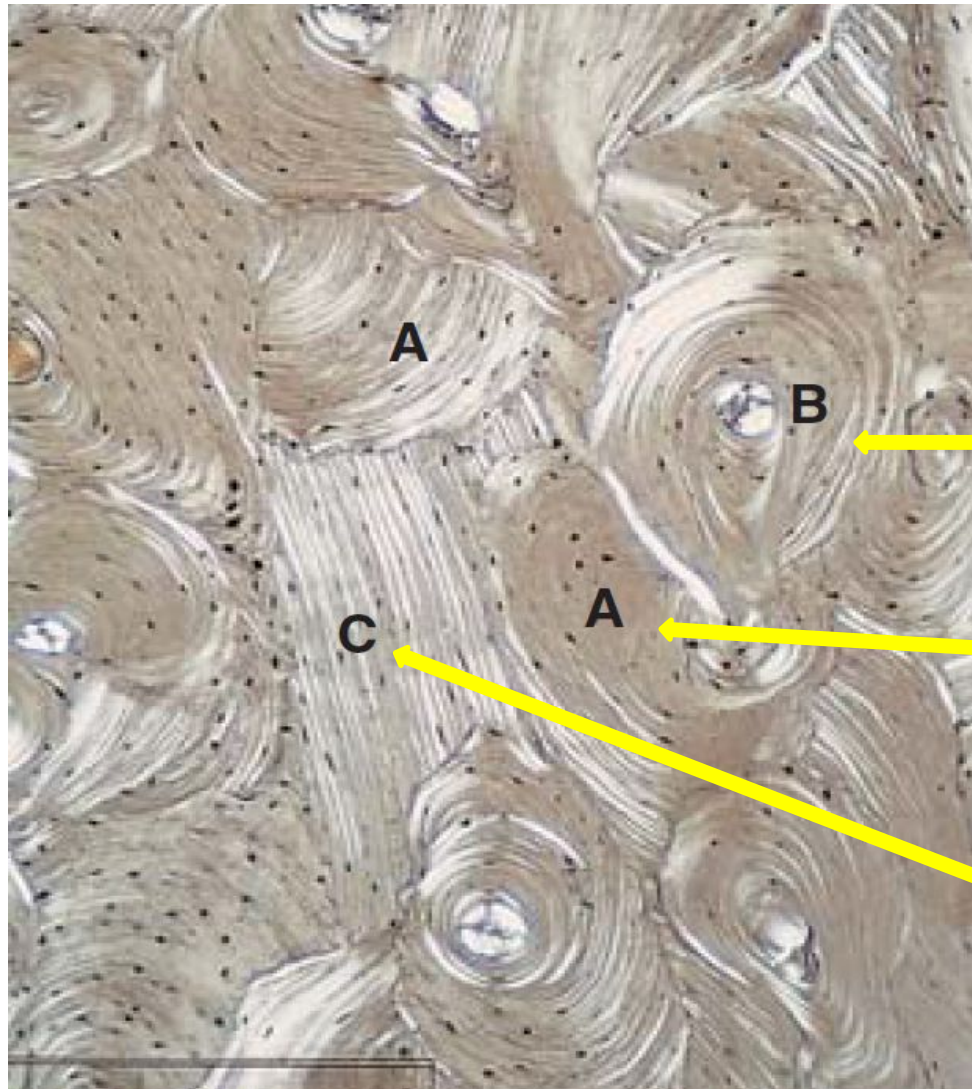


Fig. 13.5 Microradiograph of vascular canals in the cribriform plate



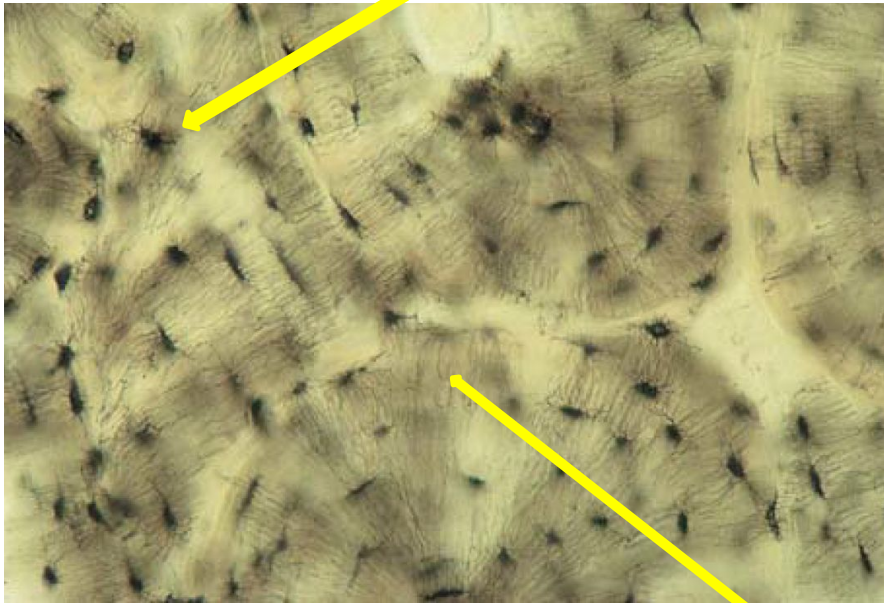
Haversian System with Central neurovascular canal

Interstitial Lamella

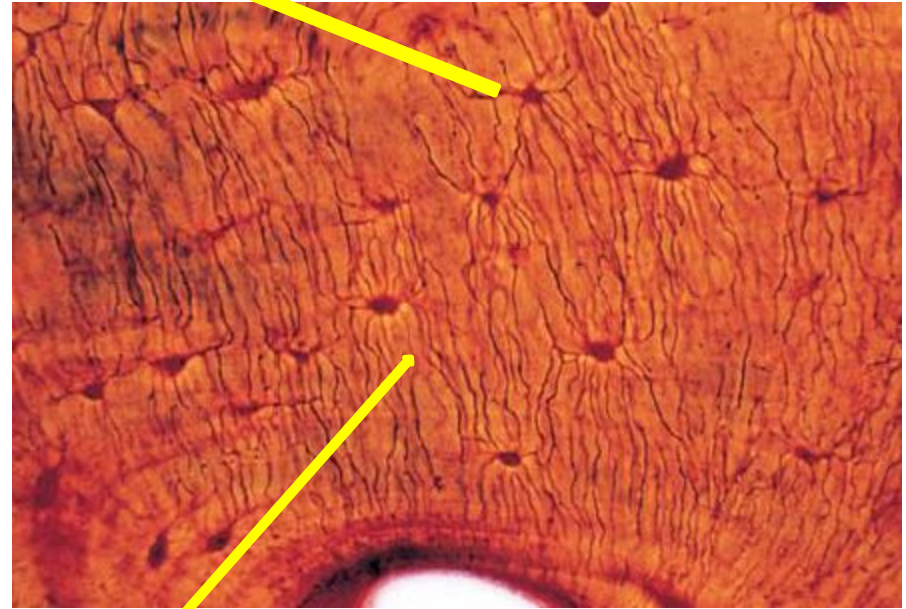
Original Circumferential lamella lying deep within bone following Remodelling

Fig. 13.8 Horizontal ground section of bone

Osteocyte lacunae



Associated canaliculi



**Fig. 13.12** Ground section of part of a Haversian System

**Fig. 13.13** Demineralized section of bone to visualize the osteocyte lacunae and canaliculi



circumferential lamellae?  
concentric lamellae?

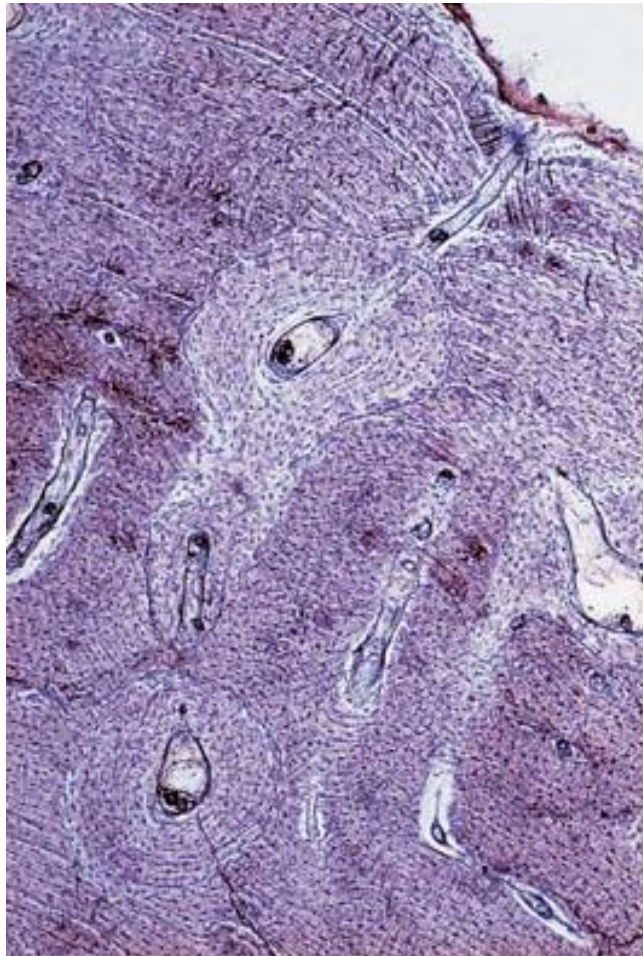


Fig. 13.6 Horizontal ground section of alveolar bone showing circumferential lamellae



Fig. 13.7 Same section as Fig. 13.6, in polarized light. Alternating black and white bands indicate different orientations of collagen in adjacent lamellae.

# Enumerate cellular components of Alveolar bone?

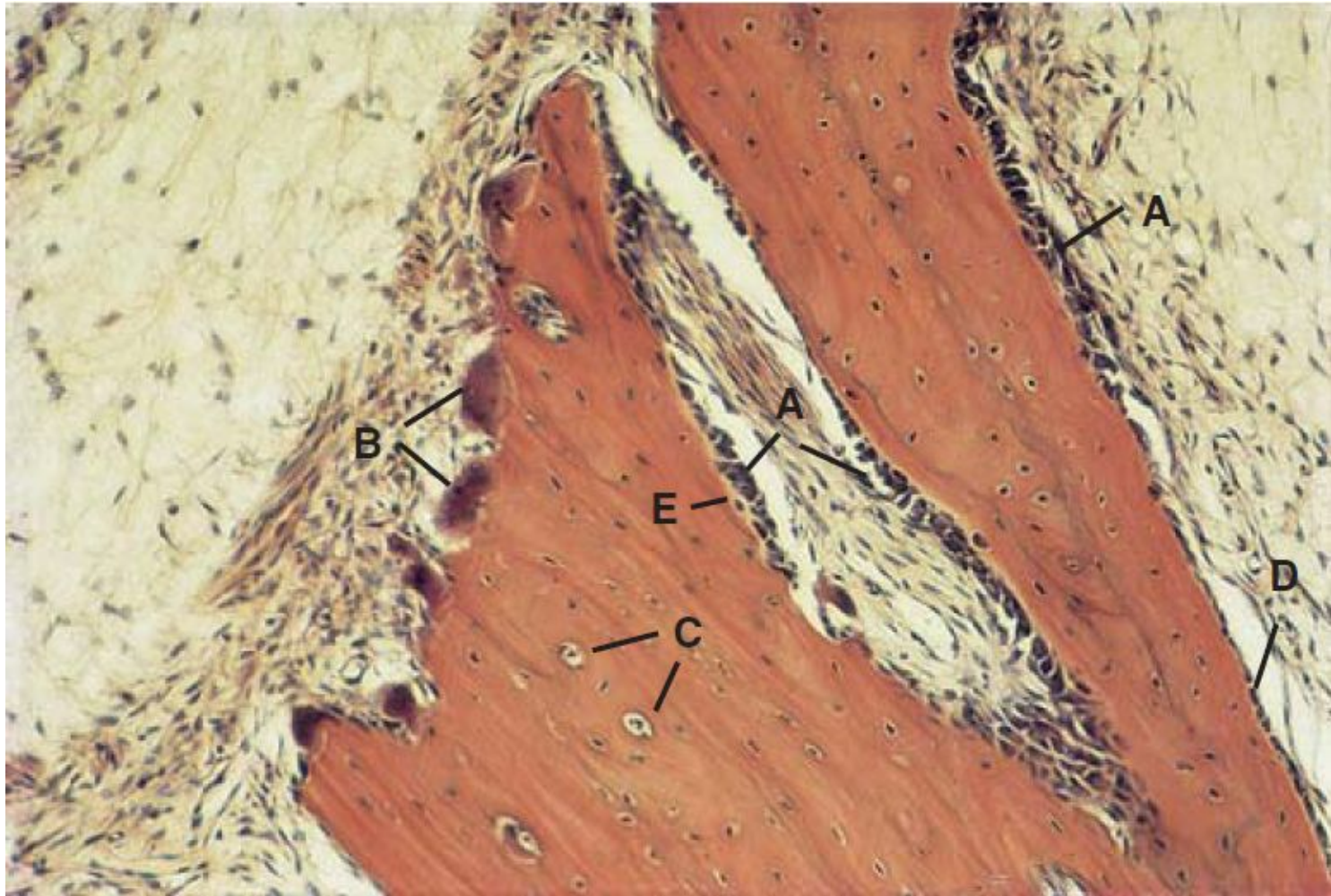


Fig. 13.9.Horizontal section of bone



A layer of osteoblasts (A) lining a surface where active bone formation occurs with the presence of a pale staining layer of osteoid),

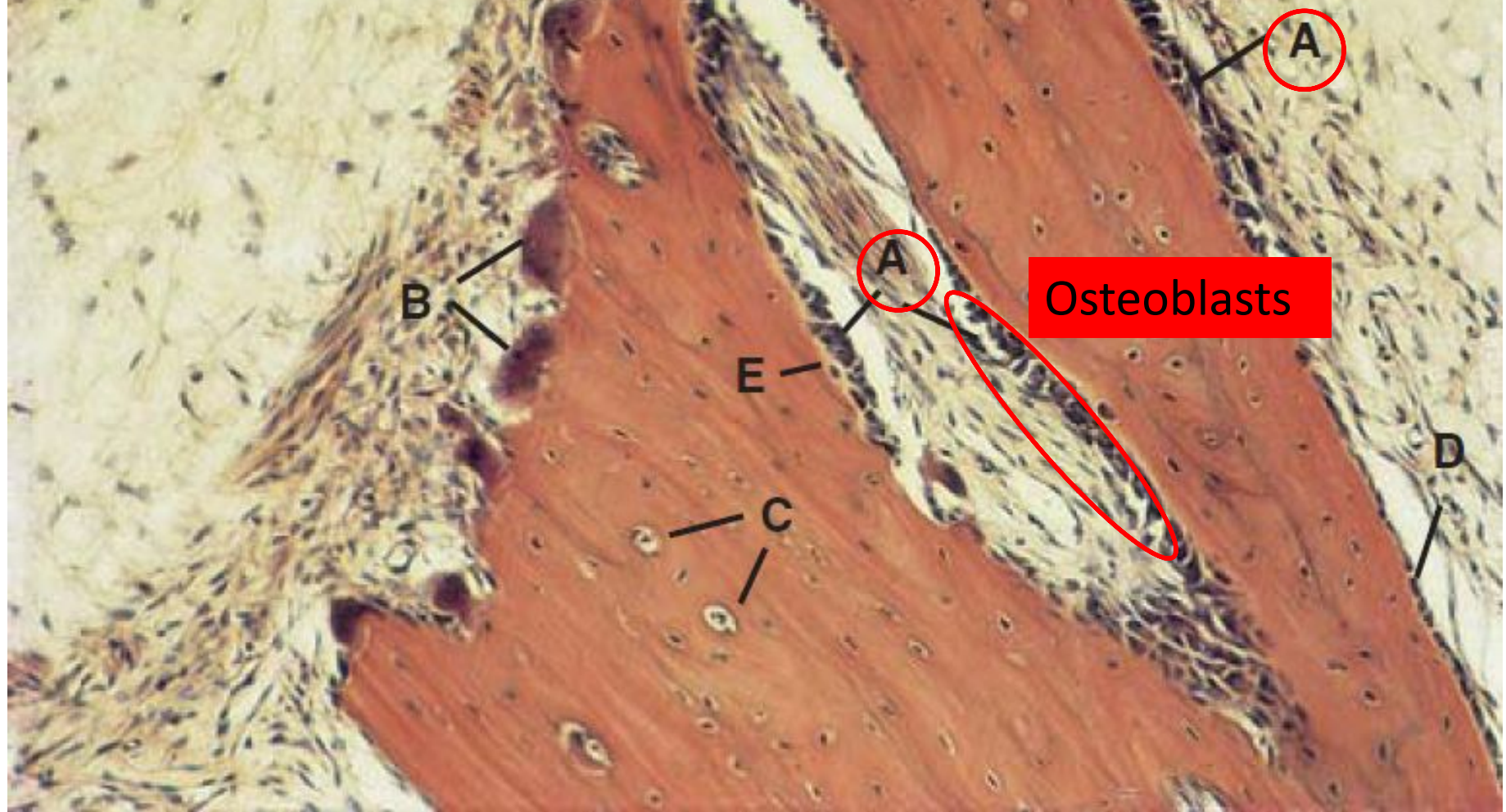


Fig. 13.9 Horizontal section of bone



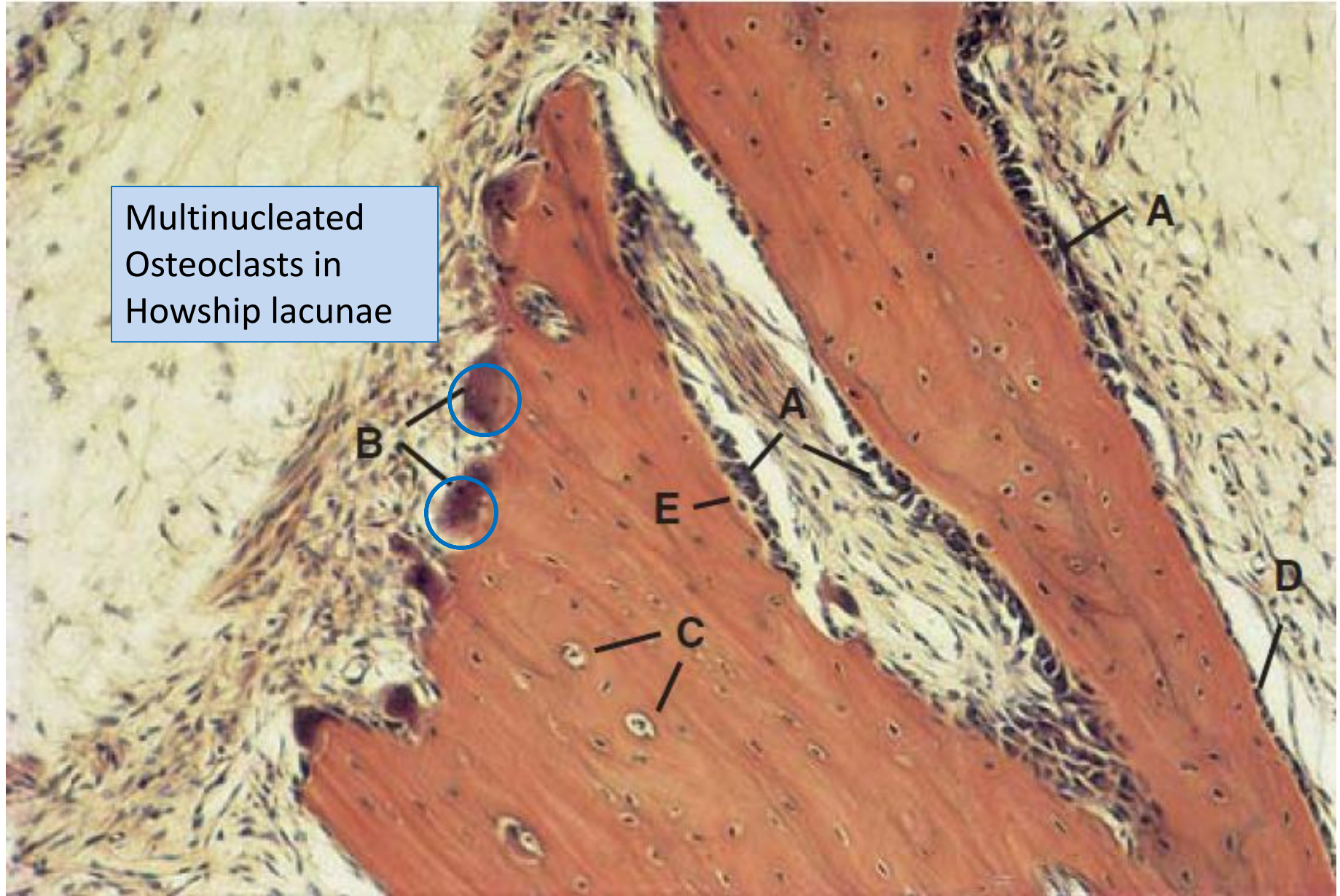


Fig. 13.9 Horizontal section of bone



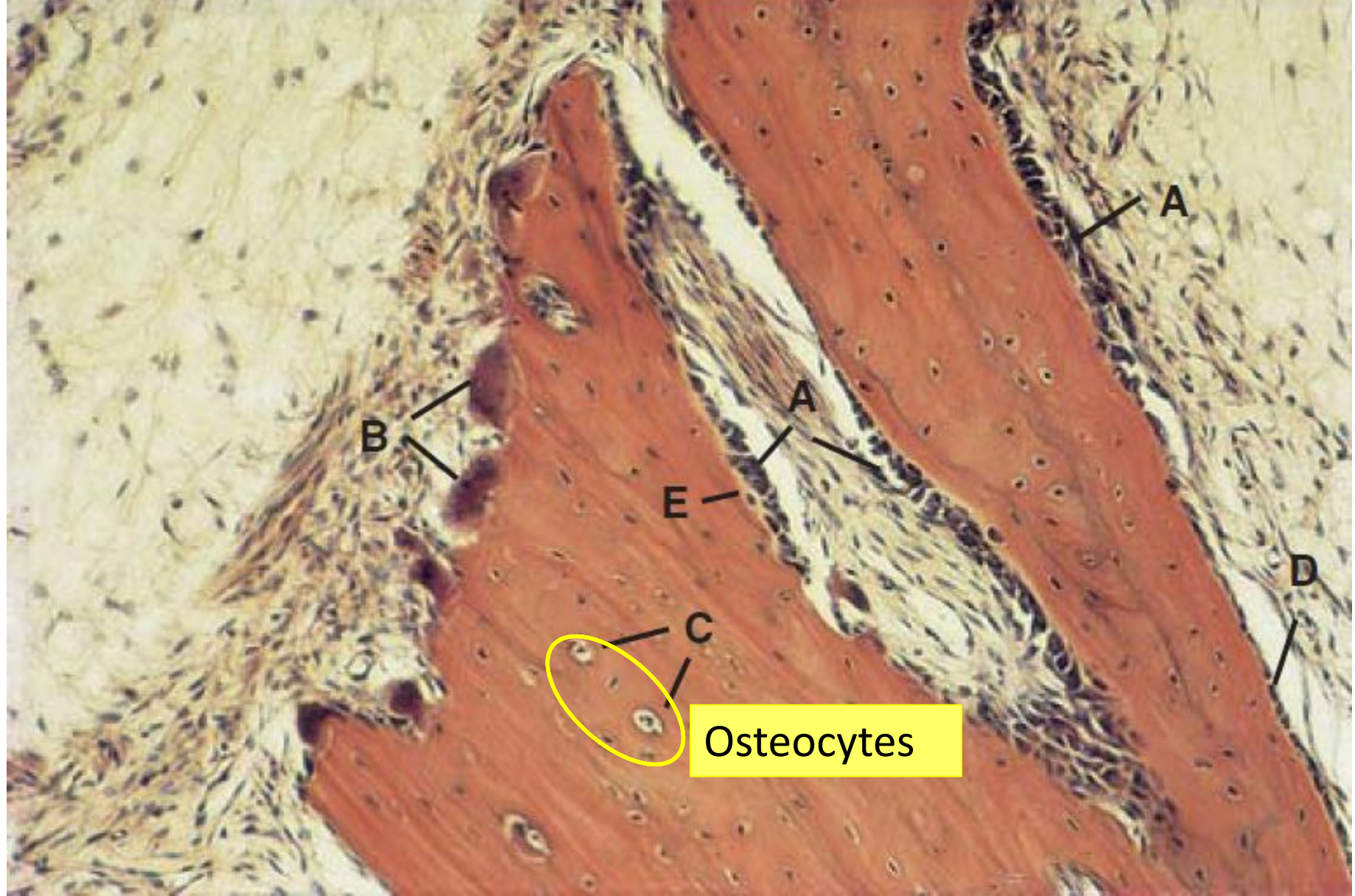


Fig. 13.9 Horizontal section of bone



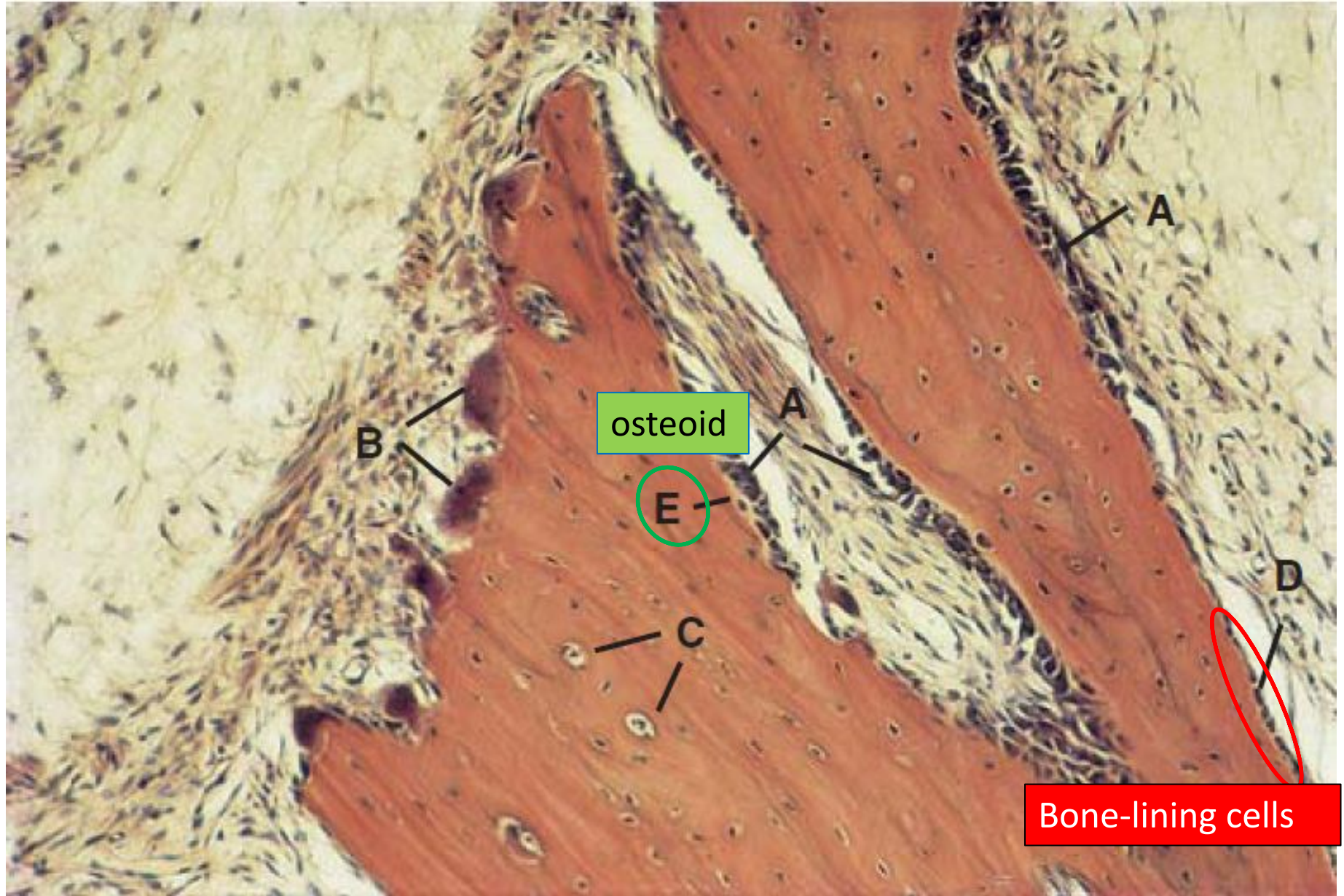


Fig. 13.9 Horizontal section of bone



A layer of osteoblasts (A) lining a surface where active bone formation is occurring (as indicated by the presence of a pale staining layer of osteoid),

Multinucleated Osteoclasts in Howship lacunae

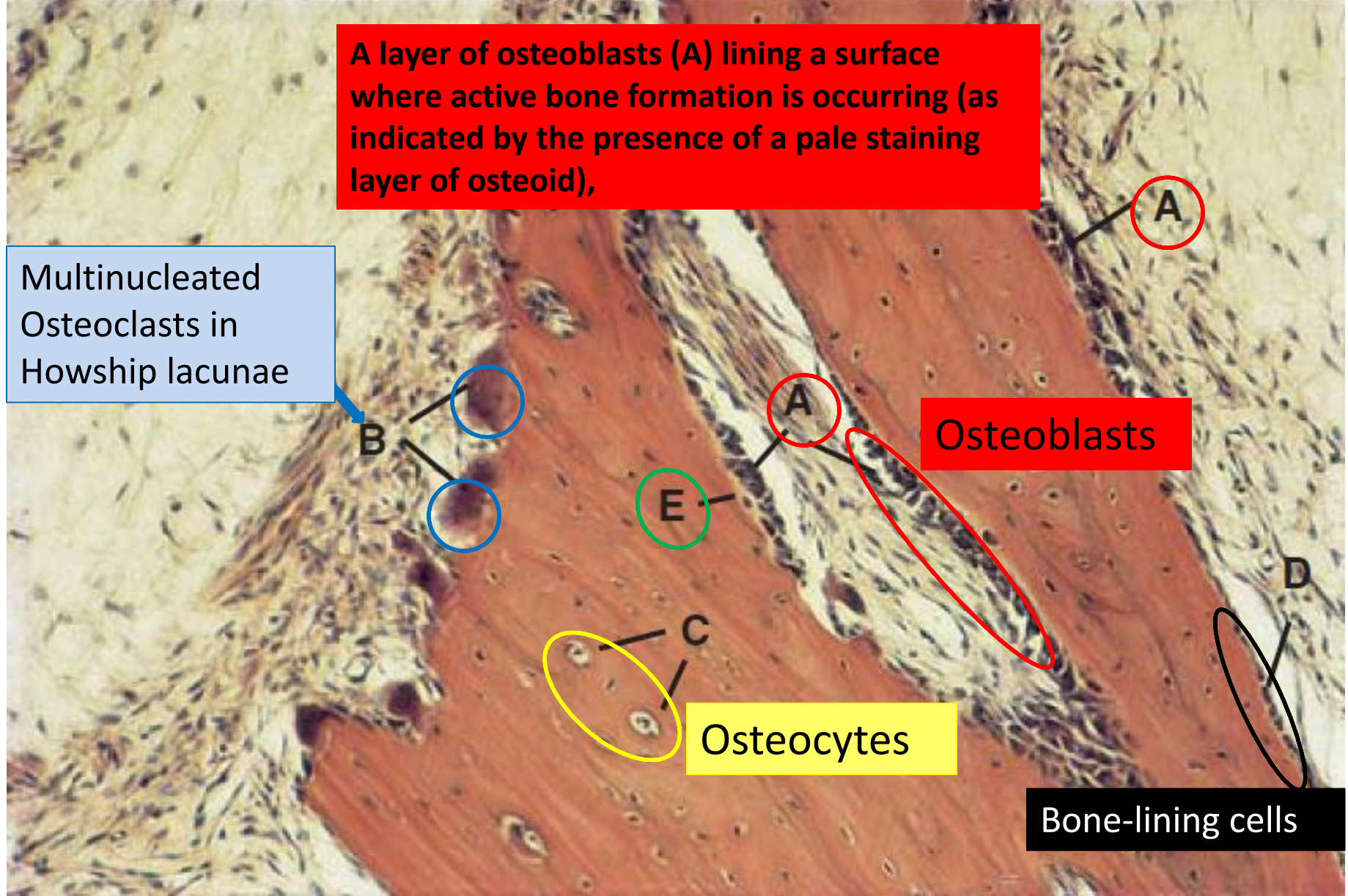
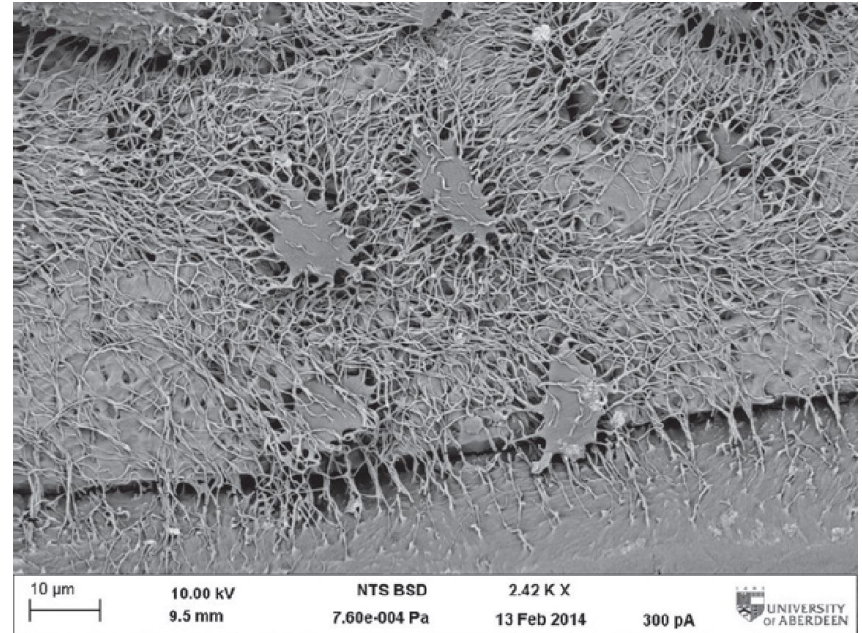
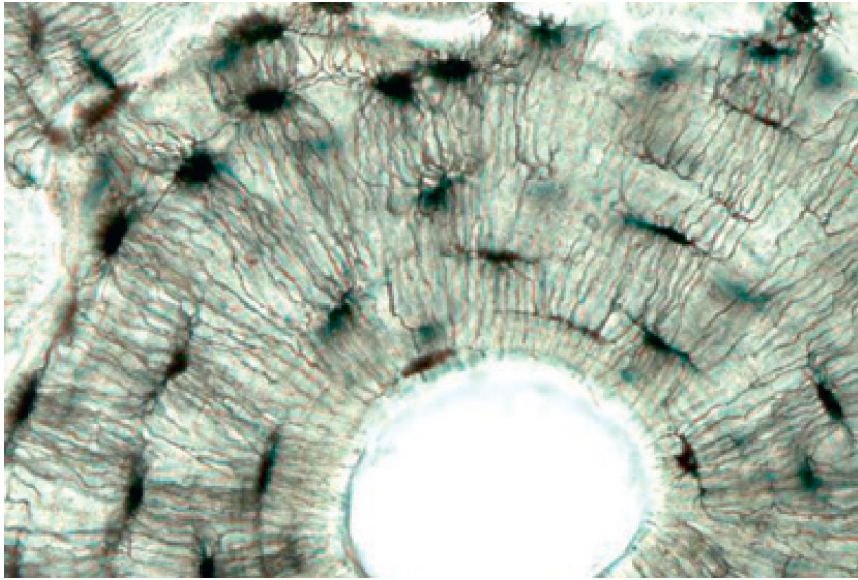


Fig. 13.9 Horizontal section of bone



**Fig. 13.12.** Haversian system cut horizontally to show osteocyte lacunae and associated canaliculi

# **Differentiate between osteoblast and osteoclast in reference with origin, composition, structure and function.**

## **OSTEOCLASTS**

- Haemopoietic in origin
- Multinucleated giant cells that resorb bone
- Motile
- Specialized organ of resorption (ruffled border) adjacent to bone surface
- Rich in acid phosphatase.
- Ruffled border pumps acid (via  $H^+$ -ATPase) to dissolve bone mineral and
- enzymes (mainly cathepsin K) to degrade collagenous matrix
- Early expression of the nuclear transcription factor c-Fos.

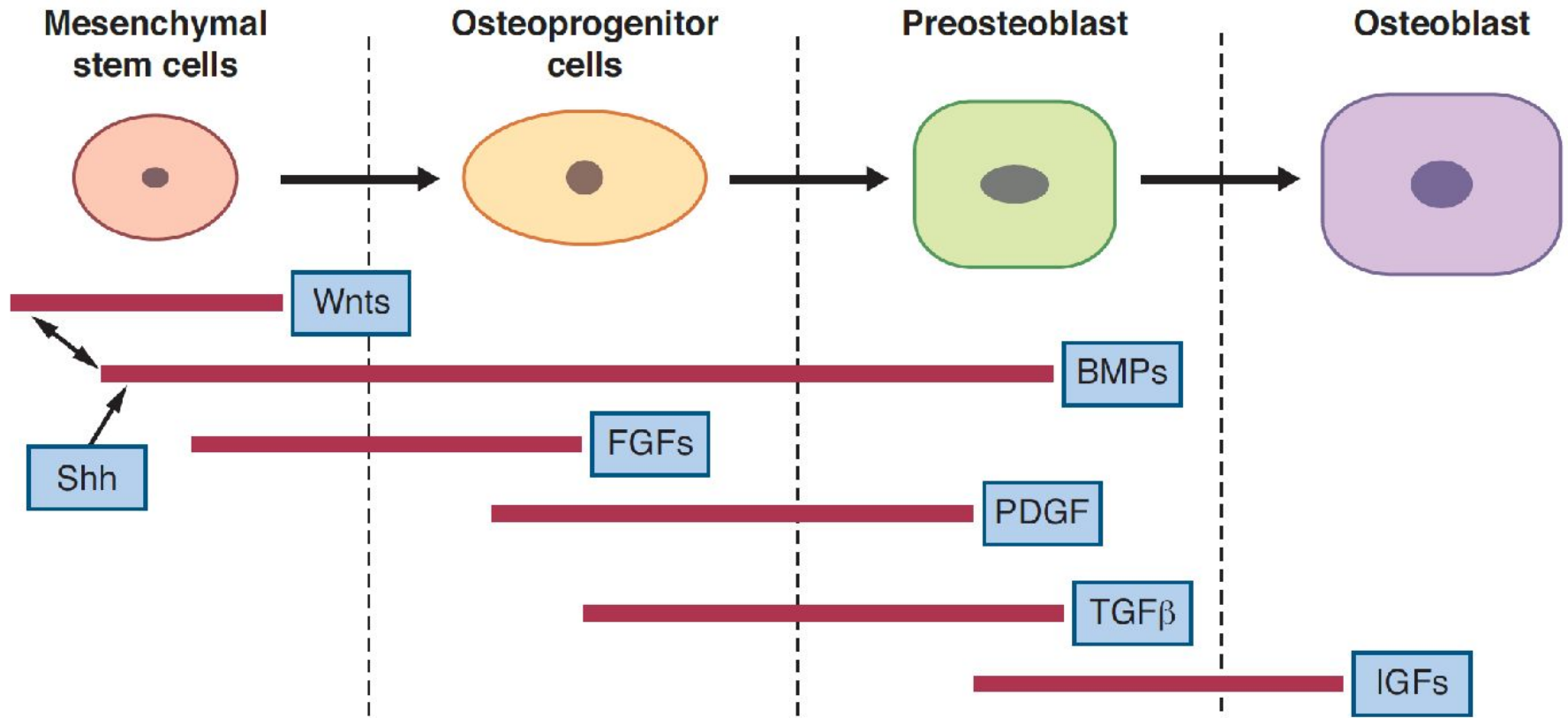


## Osteoblasts

- Mesenchymal in origin
- Lie on surface of forming bone
- Secrete and mineralize organic matrix
- Incorporated into bone matrix as osteocytes
- Rich in alkaline phosphatase.
- Early expression of the nuclear transcription factor core binding factor 1
- Cbfa1 (also called Runx2)
- Rich in endoplasmic reticulum and Golgi apparatus

## Osteocytes

- Mesenchymal in origin
- Lie within bone and surrounded by bone matrix
- Most abundant cells in bone
- Communicate with canaliculi
- Fewer intracellular organelles compared with osteoblasts
- May prevent hypermineralisation by releasing pyrophosphate
- Endocrine/paracrine regulators of bone and mineral metabolism (FGF23, Sclerostin, RANK ligand)



**Fig. 13.26** The main stages of growth factor activity during osteoblast differentiation.

BMPs = bone morphogenetic proteins; FGFs = fibroblast growth factors; IGFs = insulin-like growth factors; PDGF = platelet-derived growth factor; Shh = sonic hedgehog; TGF $\beta$  = transforming growth factor- $\beta$ ; Wnts = Wnt proteins.

# Resorption and Formation of Bone during Remodelling

- The processes of bone resorption and formation at sites of remodeling do not occur randomly and balance between the two processes is referred to as coupling.
- From a resting state, the sequence of remodeling consists of four main phases that occur in a remodeling cycle:
- 1. **resorption**: recruitment, migration and activation of osteoclasts causing bone resorption (Fig. 13.30A)
- 2. **reversal**: cessation of resorption and disappearance of osteoclasts by apoptosis or migration; the site becomes occupied by mononuclear cells (Fig. 13.30B)
- 3. **formation**: osteoblast recruitment, migration, differentiation and formation of new bone in the resorption site (Fig. 13.30C)
- 4. **resting**: formation of bone ceases following mineralisation and the surface is lined by a flattened layer of cells

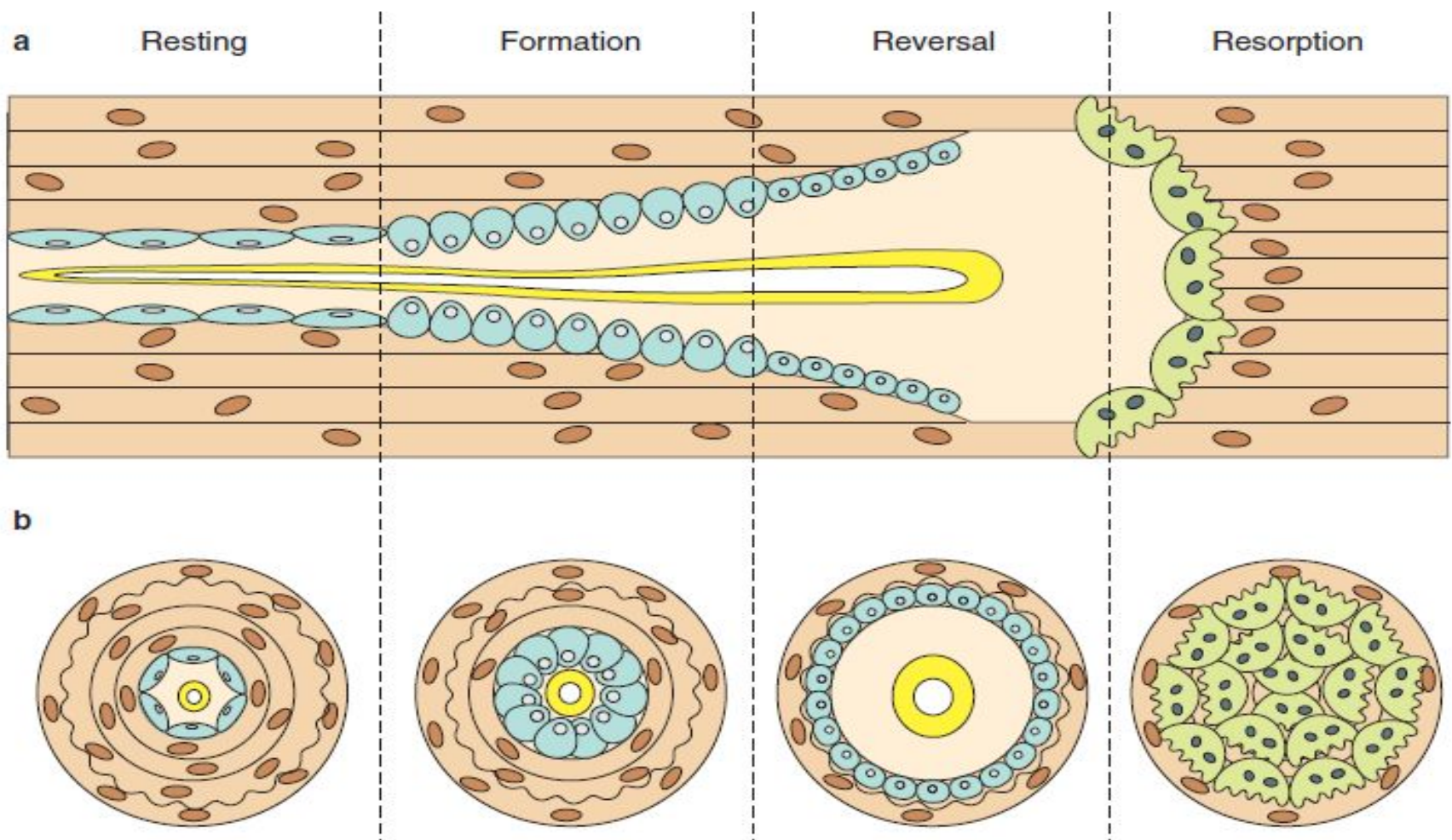


Fig. 13.26 A cutting cone.

(a) Longitudinal section along a haversian system showing the four main phases of the remodelling cycle.

(b) Cross-sections of the haversian system showing the corresponding appearance of each of the four main phases of the remodelling cycle.



| Table 13.2 Key natural regulators of bone remodelling and mineral metabolism |                                                                                |                                              |
|------------------------------------------------------------------------------|--------------------------------------------------------------------------------|----------------------------------------------|
| Regulator                                                                    | Action                                                                         | Produced by                                  |
| <b>Systematic hormones</b>                                                   |                                                                                |                                              |
| Parathyroid hormone                                                          | ↑ blood calcium<br>↑ bone resorption<br>↑ bone formation                       | parathyroid glands                           |
| 1,25 dihydroxyvitamin D                                                      | ↑ blood calcium & mineralisation<br>↑ bone resorption                          | skin (UV light)/liver/kidney                 |
| Glucocorticoids                                                              | ↓ bone formation                                                               | adrenal glands                               |
| Oestrogen (& testosterone)                                                   | ↓ bone resorption<br>↑ bone formation (smaller effect)                         | ovaries (& testes); adrenals                 |
| <b>Locally-acting factors</b>                                                |                                                                                |                                              |
| RANK ligand (RANKL)                                                          | ↑↑ bone resorption (forms osteoclasts)                                         | osteoblasts; osteocytes                      |
| OPG                                                                          | ↓ bone resorption (blocks RANKL)                                               | osteoblasts; osteocytes                      |
| Inflammatory cytokines (interleukins 1 & 6; TNFα)                            | ↑ bone resorption                                                              | osteoblasts; immune cells                    |
| Sclerostin                                                                   | ↓ bone formation (blocks Wnt signalling)                                       | osteocytes                                   |
| BMPs                                                                         | ↑ bone formation                                                               | osteoblasts; mesenchymal cells               |
| VEGF                                                                         | ↑ bone formation (via growth of new blood vessels)                             | osteoblasts; many other cells                |
| <b>'Fundamental' (Inorganic) agents</b>                                      |                                                                                |                                              |
| Nitric oxide                                                                 | ↑ bone formation (dilates blood vessels)                                       | blood vessel lining cells; nerves            |
| Hypoxia (low oxygen)                                                         | ↑ bone resorption (forms osteoclasts)<br>↓ bone formation (blocks osteoblasts) | ischaemia                                    |
| Acidosis (low pH)                                                            | ↑ bone resorption (activates osteoclasts)                                      | kidney disease; tumours; diabetes; ischaemia |
| Pyrophosphate                                                                | ↓ bone mineralisation                                                          | breakdown of ATP                             |
| <b>Mechanical</b>                                                            |                                                                                |                                              |
| Strain/deformation                                                           | ↑ bone formation                                                               | mechanical loading                           |

**Stem:** In bone biology, the continuous remodeling process is essential for maintaining bone integrity.

**Lead-in:** The primary structural unit of bone tissue that contributes to its remodeling throughout life is:

1. Osteoblast
2. Osteoclast
3. Haversian system
4. Periosteum
5. Endosteum

- **Answer:** 3. Haversian system

**Stem:** Bone formation can occur through different processes.

**Lead-in:** The type of bone formation that occurs without a cartilaginous model, forming directly within a vascular connective tissue, is:

1. Endochondral bone formation
2. Intramembranous bone formation
3. Periosteal bone formation
4. Endosteal bone formation
5. Compact bone formation



- **Answer: 2. Intramembranous bone formation**

**Stem:** Bone remodeling is a dynamic process involving various cell types.

**Lead-in:** The cells responsible for detecting mechanical stresses and strains to which the bone is subjected are:

1. Osteoblasts
2. Osteoclasts
3. Osteocytes
4. Osteoprogenitor cells
5. Chondrocytes

- **Answer: 3. Osteocytes**

**Stem:** Bone can exist in different forms during various stages of development and repair.

**Lead-in:** One distinguishing characteristic of woven bone compared to mature adult lamellar bone is:

1. Smaller and regularly spaced osteocytes
2. Regularly arranged collagen fibers
3. Higher turnover rate and rapid formation
4. Delayed mineralization
5. Lower volume of osteocytes



- **Answer:** 3. Higher turnover rate and rapid formation